

# SYSTEMS THINKING

## Systems Analysis & Design

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# Outline

1 Introduction to Systems Thinking

2 Systems Properties

3 Systems Classification



# Outline

- 1 Introduction to Systems Thinking
- 2 Systems Properties
- 3 Systems Classification



# Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.

• Not all elements need to be connected to each other, but they should be connected to the system.

• The more elements there are, the more complex the system becomes. The representation must be feasible.

- Each element must have at least one connection. Isolated elements make no sense in a system.

No

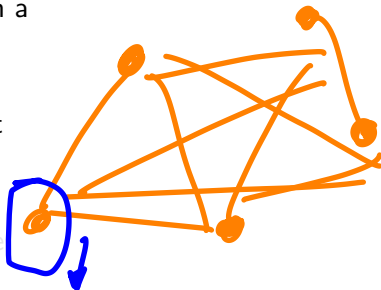


Interested



# Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.
- Not all elements need to be connected to each other, but every connection should be meaningful.
- The more connections there are, the more complex the system becomes. The representation must be feasible.
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# Introduction to Systems Thinking I

- A **system** is a set of **interconnected elements** with a **common purpose**.
- Not all elements need to be connected to each other, but every **connection** should be **meaningful**.
- The more **connections** there are, the more **complex** the system becomes. The representation must be **feasible**.
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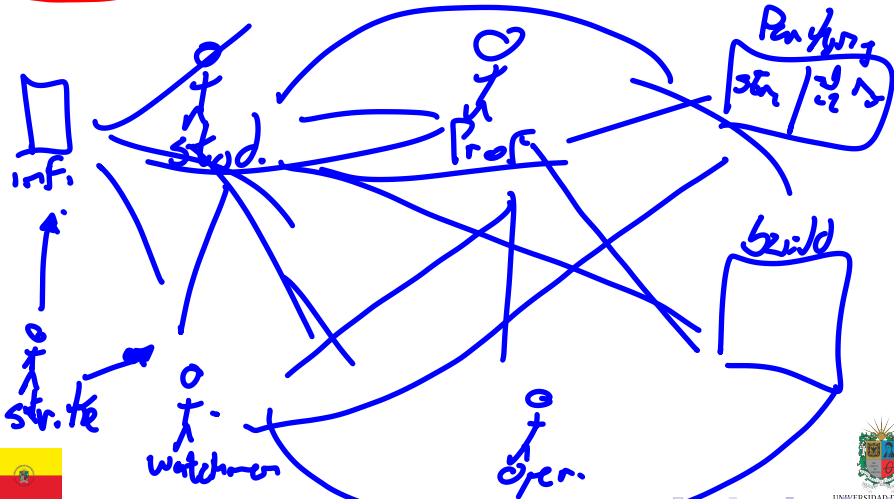
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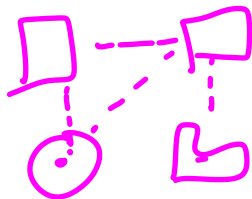


# Complexity in Systems

**System complexity** can be defined as the number of elements and connections in a system.



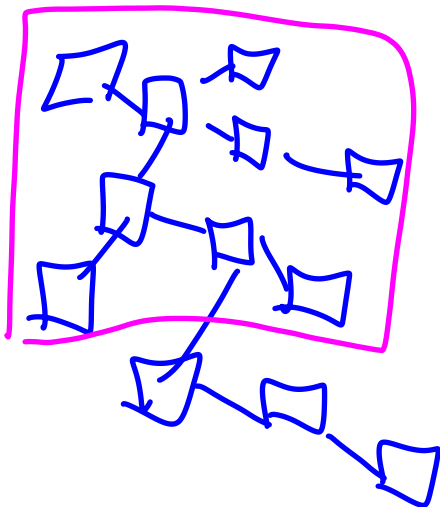
# Introduction to Systems Thinking II



- In **systems thinking**, if you just **split** the parts and forget the relationships, you will **lose the full picture**.
- It is called a **holistic** approach; try to see **the whole picture** with all the meaningful details.
- **Defining the box boundaries** is sometimes tricky; as we said, not too complex, not too simple. It is like the desired universe **balance** of Thanos.



# Introduction to Systems Thinking II



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# Introduction to Systems Thinking III

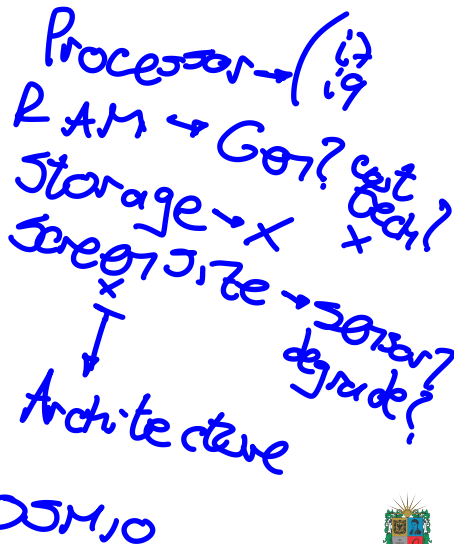
- Another important concept is **homeostasis**; it means putting a system in an equilibrium state. That is hard because systems are both not in equilibrium and resilient to change.

- A **system** is more than the sum of its parts. This means relationships, behaviors, and recovery capacity are **forgotten** when you see the system just as its parts.



# Introduction to Systems Thinking III

- Another important concept is **homeostasis**; it means putting a system in an **equilibrium** state. That is hard because **systems** are both **not in equilibrium** and **resilient to change**.
- A **system** is more than the **sum of its parts**. This means **relationships**, **behaviors**, and **recovery capacity** are **forgotten** when you see the system just as its parts.



# Introduction to Systems Thinking IV

- **Systems thinking** is a way to understand and represent problems in order to find the best possible solution.
- Thinking of a **problem as a system** lets you understand details, involved elements, and relevant information.

computation



TM. → Jst (BLT) 3

metro - subway



# Introduction to Systems Thinking IV

- **Systems thinking** is a way to **understand** and **represent** problems in order to find the best possible **solution**.
- Thinking of a **problem** as a **system** lets you understand **details**, **involved elements**, and relevant **information**.
- **Systems** should be **viable**, **self-sustainable**, provide internal **feedback loops**, and also look like a **whole living entity**.

2024 ~ 4 billion USD  
2025 ~ 11 billion USD

Cultural

OpenAI

Art. ficial Int.  
Artificial Life

cybernetics



## Introduction to Systems Thinking V

Automatic  
calculus

Babbage

Boole

maths → algorithm → code → automation  
 & robots

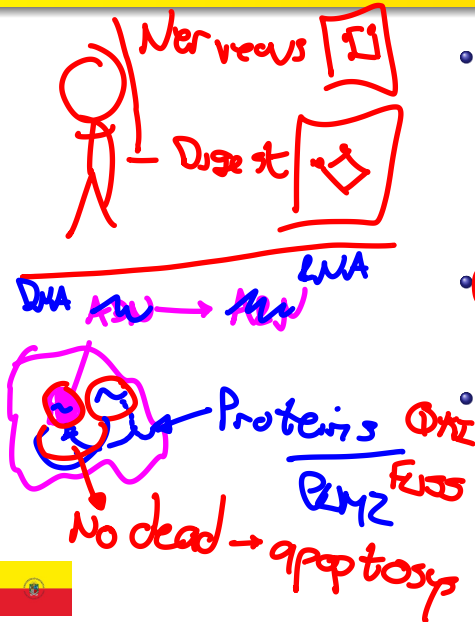
- **Computation** helps to represent behaviors mathematically. Also, it helps to find patterns and information, and simplify processes; an example of all this is Artificial Intelligence.

LLM → Next word } m./10<sup>7</sup>  
 5./10<sup>9</sup>

- The **Bottom-Up** approach is useful when you want to see the parts and then put them together to get the full picture.



# Introduction to Systems Thinking V



- **Computation** helps to **represent behaviors** mathematically. Also, it helps to **find patterns** and information, and simplify processes; an example of all this is **Artificial Intelligence**.
- **Top-Down** approach is useful when you want to see the full picture, and then split it into parts. ✓
- The **Bottom-Up** approach is useful when you want to see the parts and then connect them to get the full picture. ←



# Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to better decision-making.

- The simplest system definition

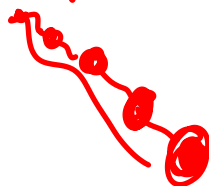
is: given some inputs, after applying a designed process to them, you will obtain some outputs.



- In a deterministic world, the same inputs produce the same outputs. However, real life is not deterministic.

Control

Snowball



# Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to better decision-making.
- The simplest **system definition** is: given some **inputs** after applying a designed **process** to them, you will obtain some **outputs**.
- In a **deterministic** world, the same **inputs** produce the same **outputs**. However, real life is **not deterministic**.

Engineering

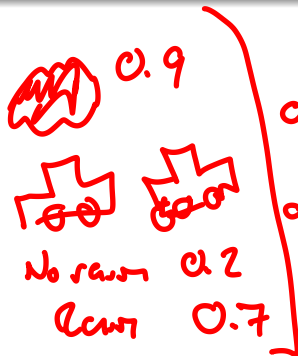
$$\begin{array}{r} I \quad G \\ 2+2=4 \\ 3+3=6 \end{array}$$

controlled system

~ TC



# Introduction to Systems Thinking VII



$$0.9 \times 0.7$$

$$0.1 \times 0.2$$

Bayes Theorem

- Since **randomness** is normal in the real world, relying solely on deterministic processes is **dangerous**. Using **stochastic processes** is a better approach.
- **Stochastic processes** make use of **probability** which leads to a better representation of real-world behavior.

Here, Chaos Theory becomes a useful tool. In fact, simply, chaos can be defined as a harmonious balance between rules and randomness.



# Introduction to Systems Thinking VII

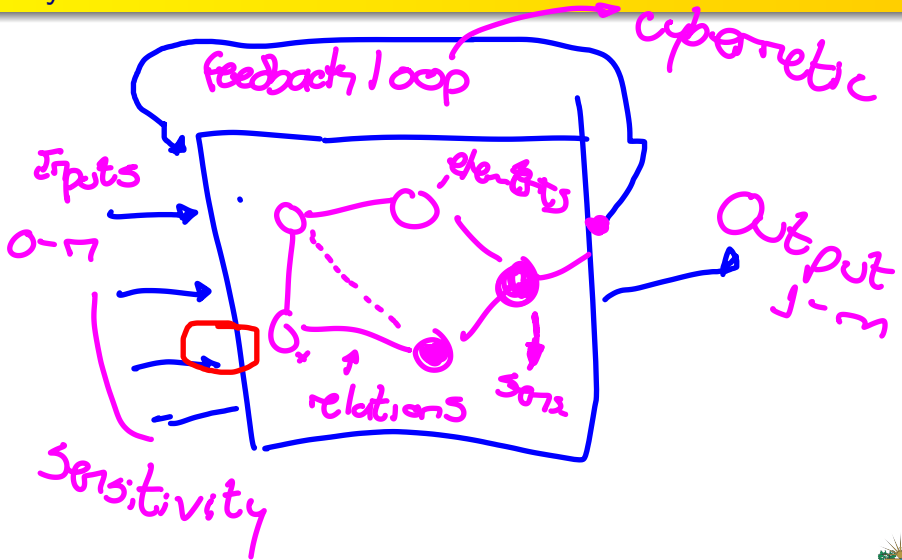


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Chaos



# Systems Structure Draw



# Case of Study: Transportation System



# Outline

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- 2 Systems Properties
- 3 Systems Classification



# Systems Properties I

- **Emergence** is a property of systems that means that the whole system is more than the sum of its parts.

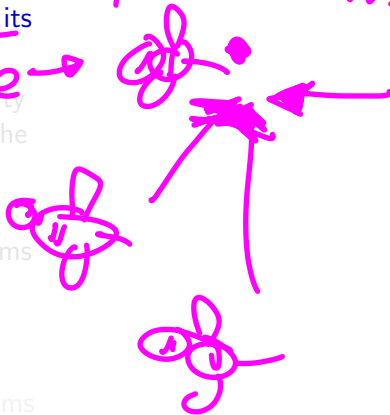
- **Interconnectedness** is a property of systems that means that all the elements are connected in a meaningful way.

- **Feedback** is a property of systems that means that the system has internal loops that control the system behavior.

- **Hierarchy** is a property of systems that means that the system has levels of organization.

arise from system evolving

bee →



time  
↓  
simulations



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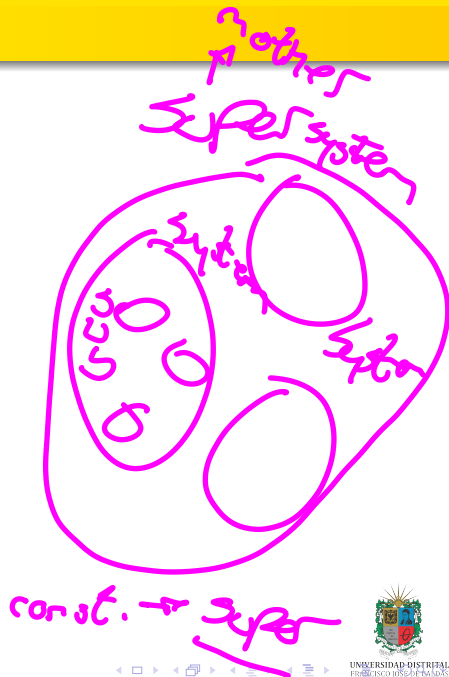
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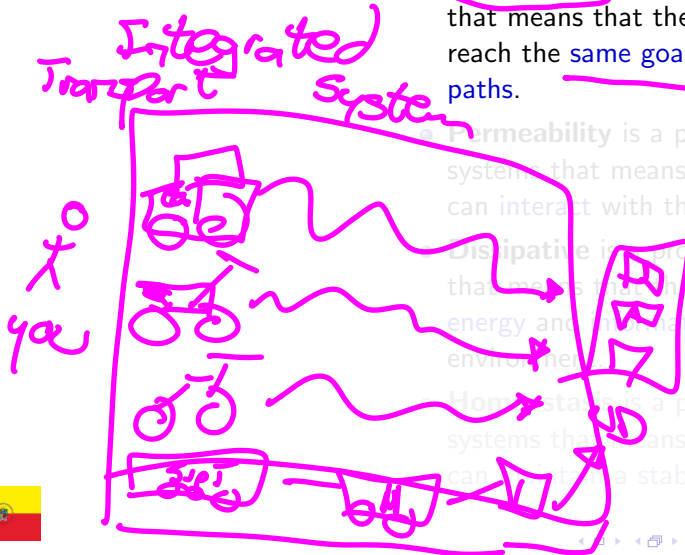
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# Systems Properties II

- **Equifinality** is a property of systems that means that the system can reach the **same goal** from **different paths**.

- **Permeability** is a property of systems that means that the system can **interact with the environment**.
- **Dissipative** is a property of systems that means that the system can **lose energy and information** to the environment.
- **Homeostasis** is a property of systems that means that the system can **maintain a stable state**.



# Systems Properties II

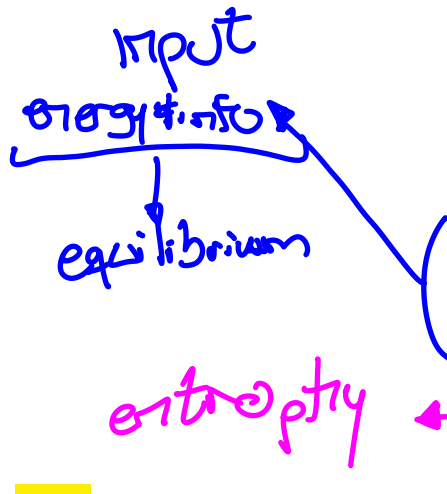


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# Systems Properties III

- **Equilibrium** is a property of systems that means that the system can reach a stable state

→ *sustaining*

- **Adaptability** is a property of systems that means that the system can *change* to *adapt* to the environment.
- **Self-organization** is a property of systems that means that the system can *organize* itself.
- **Self-regulation** is a property of systems that means that the system can *regulate* itself.



# Systems Properties III

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• **Self-organization** is a property of systems that means that the system can organize itself.

• **Self-regulation** is a property of systems that means that the system can regulate itself.

mutation → fitness

evolution

resilience to change



generation 5

orderly



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● **Self-regulation** is a property of systems that means that the system can **regulate** itself.

AIDS



Tardigrade

*Hand-drawn scribbles*



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→ sustainability



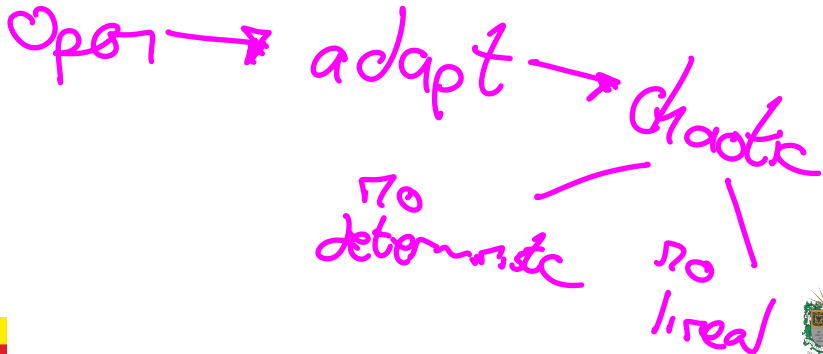
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# Basic Concepts

A **system** can be classified according to different criteria like openness, adaptability, determinism and linearity.



# Systems Classification I

- **Open systems** are systems that can interact with their environment.

} almost all

- **Closed systems** are systems that cannot interact with their environment.

- **Adaptive systems** are systems that can change and adapt to their environment.

} laboratory experiments

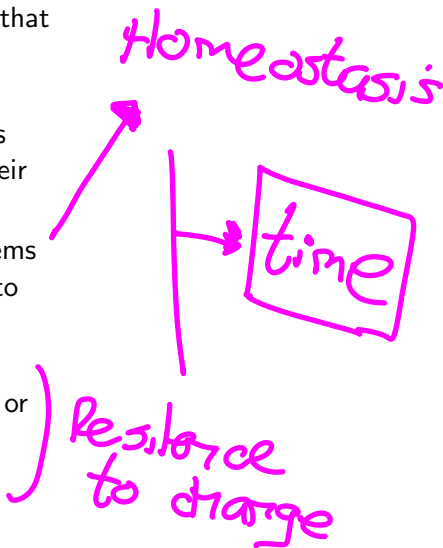
- **Non-adaptive systems** are systems that cannot change or adapt to their environment.

} highly controlled



# Systems Classification I

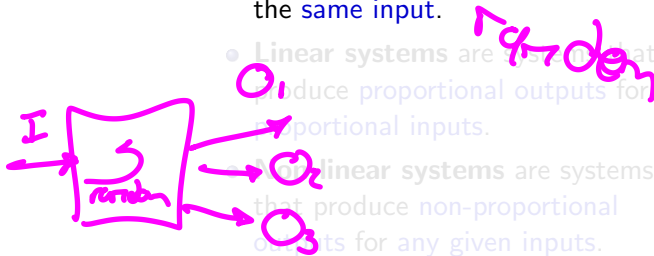
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# Systems Classification II



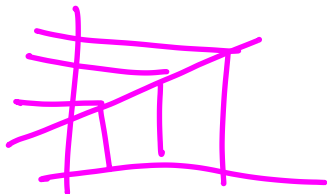
- **Deterministic systems** are systems that produce the **same output** for the **same input**.
- **Stochastic systems** are systems that produce **different outputs** for the **same input**.



- **Linear systems** are systems that produce **proportional outputs** for **proportional inputs**.
- **Non-linear systems** are systems that produce **non-proportional outputs** for any given inputs.



# Systems Classification II



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- **Linear systems** are systems that produce proportional outputs for proportional inputs.
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*chaotic & stochastic*



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# Thanks!

## Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

